Ran Kiri

PhD student in Mathematics, Technion – Israel Institute of Technology

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Education

PhD in Mathematics, Technion — in progress
Advisor: Prof. Orr Moshe Shalit

MSc in Mathematics, Technion — 2020–2022
Thesis: *Finitely Generated Matrix Convex Sets and Classification of Operator Systems Generated by Λ-Commuting Unitaries*.
Grade 95; GPA 94.3.
Advisor: Prof. Orr Moshe Shalit

BSc in Mathematics and Physics, Technion — 2016–2020
GPA 83.9

Research interests

Operator systems and their geometric structure; matrix convexity and noncommutative convex sets; Arveson's essential normality conjecture; von Neumann–type inequalities.

Publications

R. Kiri, *Subhomogeneous Operator Systems and Classification of Operator Systems Generated by Λ-Commuting Unitaries*, Integral Equations and Operator Theory **96** (2024), Article 15. — 2024
doi:10.1007/s00020-024-02765-8 arXiv:2302.04767

R. Kiri, *Finitely Generated Matrix Convex Sets and Classification of Operator Systems Generated by Λ-Commuting Unitaries*, MSc thesis, Technion. — 2022

Teaching

Lecturer, Faculty of Mathematics, Technion — 2023–present
Ordinary Differential Equations M; Fourier Series and Integral Transforms; Complex Functions A

Teaching Assistant, Faculty of Mathematics, Technion — 2018–present
Calculus 1M, 2 and 3; Ordinary Differential Equations A; Algebra B; Introduction to Functional Analysis; Real Analysis; Complex Functions and Integral Transforms

Course materials (lecture notes and recitations): rankiri93.github.io/courses

Awards and fellowships

Outstanding Lecturer Award, Technion — 2024–2025
Spring 2024, Winter 2024–25 and Spring 2025; commendation for excellence in teaching, Spring 2023

Sandor Szego Prize for Excellence in Teaching (outstanding teaching assistant), Faculty of Mathematics — 2018–2025
Awarded twelve times, Winter 2018–19 to Spring 2025

Fund for the Promotion of Excellence in Mathematics Prize — 2022–23
For an outstanding master's thesis

Miriam and Aaron Gutwirth Fellowship — PhD

Talks and posters

Noncommutative Geometric Structure of Subhomogeneous Operator Systems July 2025
IWOTA 2025, University of Twente slides

Subhomogeneous Operator Systems and Classification of Operator Systems Generated by Λ-Commuting Unitaries August 2023
IWOTA 2023, University of Helsinki slides

Subhomogeneous Operator Systems (poster) June 2023
Technion Research Day

What is a UCP Map? (invited lecture) May 2021
Technion slides

Conferences attended

Dilation and Classification in Operator Algebra Theory, University of Copenhagen October 2022

Service and outreach

Faculty committee member, Faculty of Mathematics, Technion
Organizing social events, liaising with faculty, helping new students

Volunteer editor, "Mathematical Drive" project
Collecting and organizing study materials for undergraduates

Military service

Reserve duty, Israel Defense Forces 2023–present
Volunteered for 199 days of reserve duty as a tank operator and tank commander during the Iron Swords War